

Test 1

1

a) Show how to calculate $C(11, 3) = \binom{11}{3}$ = **without a calculator. Show your work.**

b) Write a counting problem for which $\binom{11}{3}$ gives the answer. (Keep your problem simple.)

2 A bagel shop has 6 types of bagels. How many ways are there to select 9 bagels?

3 Bitstrings

- a) How many bitstrings of length 8 are there?
- b) How many bitstrings of length 8 have *exactly* 5 1's?
- c) How many bitstrings of length 8 have *at least* 5 1's?
- d) How many bitstrings of length 8 have at least 5 *consecutive* 1's?

4 Below is an adjacency matrix representation of a graph G .

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

a) How many vertices does G have? How can you tell?

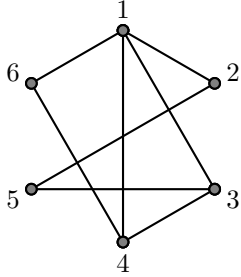
b) How many edges does G have? How can you tell?

c) Is G directed or undirected? How can you tell?

d) Does G have any self-loops? How can you tell?

e) Does G have any multi-edges? How can you tell?

5 Answer the following questions about the graph below.



- a) What is an Euler circuit? Does this graph have an Euler circuit? How do you know?
- b) Does the graph have an Euler path that is not a circuit? How do you know?
- c) What is a Hamilton circuit? Does this graph have a Hamilton circuit? How do you know?
- d) What is a bipartite graph? Is this graph bipartite? How do you know?
- e) What is a planar graph? Is this graph planar? How do you know?

6 For **one** of K_5 or $K_{3,3}$ (your choice),

a) Draw the graph.

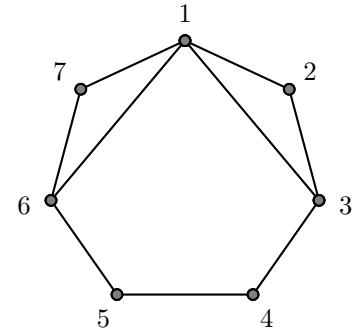
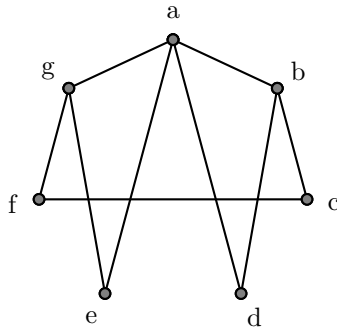
b) Explain how we know that the graph cannot be drawn in the plane with no edge crossings.

7 Draw a graph with 6 vertices, each of which has degree 3.

8 Explain why it is impossible to draw a graph with 5 vertices, all of which are degree 3.

9 How many isomorphisms are there between the following two graphs? **Explain.**

(If you do not know exactly how many there are, tell me as much as you know about how many there are. Examples: There are at least 3; there are at most 2, etc.)



Space for additional work.